

Student Name: Sarp Sürer

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Project Proposal: Quantifying the Impact of Travel Distance, Schedule Congestion, and Altitude on NBA Team Performance Metrics

Where will I get the data? The primary sports analytics dataset, encompassing game-level box scores and advanced team metrics (e.g., True Shooting Percentage, Pace, and Turnovers), will be sourced from the official NBA stats API (accessible via Python's `nba_api` package) or equivalent Kaggle historical repositories. To fulfill the mandatory data enrichment requirement, I will construct and integrate a secondary geographic and topographical dataset. This custom layer will contain the exact latitude, longitude, and elevation (altitude) of every NBA arena, sourced from public geographic databases.

How will I collect it? The entire data collection and preparation pipeline will be conducted in Python. The game-level performance data will be extracted programmatically to ensure reproducibility. The geographic arena data will be compiled as a structured CSV file. I will utilize `pandas` for advanced data manipulation and the `geopy` library to computationally calculate the exact flight distance a team traveled between consecutive away games. A key technical step will involve merging the game logs with the geographic data based on the away team's previous game location, subsequently deriving new engineered features such as "Days of Rest," "Distance Traveled in the Last 48 Hours," and "Altitude Difference."

The characteristics of the data I will be using: The synthesized dataset will be a multivariate panel dataset spanning the last three to five regular NBA seasons (excluding the COVID-19 "Bubble" season to prevent anomalies). With 1,230 games per regular season, the final dataset will contain approximately 4,000 to 5,000 individual game samples. The features will include standard advanced basketball metrics (Offensive/Defensive Ratings), engineered scheduling constraints (Back-to-back game boolean flags, cumulative rest days), and continuous geographical stressors (Flight distance in kilometers, Arena elevation in meters). This structure is designed to support exploratory data analysis (EDA), statistical hypothesis testing (e.g., evaluating the "Denver altitude effect"), and foundational machine learning regression models to predict performance dips based on schedule severity.

